

Limited-Memory Renewal Contracts: Participation, Quantization, and Exact Design

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Abstract. A renewal agreement can reach one execution state with different accepted obligations while retaining only a finite service symbol. We derive the resulting quantizer design from participation and continuation promises. In the saturated quadratic model, an optimal expected-participation codebook has cap-anchored nonhighest levels and a continuously optimized highest level that can lie strictly between caps. Contractual terminal costs remain Monge after continuous minimization, making classical matrix search applicable. Pathwise participation eliminates the saturated lottery advantage. We then characterize the full-information alphabet for every root promise and prove budget-uniform bounds between every pair of promises, with exact fixed-book allocation certificates. Strict saturated institutional advantages persist on explicit nonsaturated intervals. For arbitrary finite certification catalogs, a second exact dynamic program jointly selects codewords, bounded-overflow lotteries, and additive charges for the actual chosen levels. An example shows why pricing a previously loss-minimizing codebook is insufficient. We distinguish these contractual results from classical scalar quantization, unbiased stochastic rounding, dual quantization, and finite-rate control. Exact tests, independent support enumeration and search substitution, and provenance-preserving computational evidence assess the implementations.

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1 Introduction

An upstream renewal desk can know an accepted obligation that a downstream execution gateway does not retain. When only a small service symbol crosses that boundary, the provider must choose both its terminal instruction alphabet and the branch-dependent encoder. This is a quantizer-design problem, but its distortion and feasible randomization are generated by a contract: an expected continuation promise, local participation limits, and the cost of providing intermediate rather than terminal service. The question is how much service value each additional retained symbol buys, and how that value changes when participation is checked before or after the symbol is drawn.

Finite-alphabet encoding, adjacent unbiased randomization, and matrix-search acceleration of ordered quantizer design are established ideas. In particular, Wu (1991) already obtained work linear in the number of source points per level for optimal mean-square scalar quantization; stochastic rounding and dual quantization provide the relevant unbiased encoder primitives (Crocì et al., 2022; Pagès and Wilbertz, 2012). Our contribution is not to rename those primitives. It is to derive and solve the institution-generated quantization problem, including its costly service shortfall above the largest codeword, and to show which value comparisons survive when the aggregate promise is relaxed.

The analysis has three linked parts. First, for a saturated three-date renewal contract, deterministic execution gives an ordered one-sided quantizer, whereas expected participation permits adjacent unbiased lotteries. For heterogeneous quadratic costs, an optimal expected-participation book can be chosen with all nonhighest levels at branch caps and one continuously optimized highest level. The latter can lie strictly below the largest target and strictly between caps because intermediate service supplies the remaining obligation. A crossing-difference identity proves that the terminal array remains Monge after this continuous optimization. Classical favorable-staircase completion and SMAWK then yield linear exact-rational oracle and comparison work per dynamic-programming layer. Pathwise participation eliminates the lottery advantage at the saturated endpoint for every alphabet budget.

Second, saturation does not confine the decision insight to a single root promise. We characterize the full-information service levels and minimum exact alphabet at every promise by a clipped common level. A same-alphabet contraction and partial-lift argument bounds the change of every finite-budget value between any two feasible root promises uniformly. Every strict expected-versus-pathwise advantage therefore persists on a certified interval of nonsaturated promises. An exact fixed-book reduction identifies the remaining joint promise-allocation problem rather than treating branch caps as targets after they cease to bind. An exact supporting-price algorithm certifies each fixed-book allocation, including zero intermediate costs and repeated caps.

Third, we optimize bounded-overrun contracts over arbitrary finite certification catalogs with charges depending on the actual selected levels. An ordered-path dynamic program jointly chooses the codebook, service policy, and risk-feasible lotteries for every budget. This is more than pricing an already loss-minimizing book: an exact example changes the selected codewords when a single level becomes costly. The unrestricted continuous frontier and the charged finite-catalog frontier are separate, explicitly specified optimization problems.

The modeled renewal architecture is an operational abstraction, not an empirical claim about a named

service provider. We use renewal language throughout and do not infer customer risk acceptance from the architecture. Exact regression, independently authored search substitution, reduced-network solver certificates, and computational measurements assess the algorithms; none substitutes for a proof of the contract reduction. A focused electronic companion contains the supporting proofs and implementation evidence. The complete preceding readers and broader compact-response theory remain independently accessible in the repository archive.

2 Quantization, Contracting, and the Contribution Boundary

Ordered scalar quantization. Wu (1991) accelerates the discrete optimal mean-square K -level quantization dynamic program from $O(KN^2)$ to $O(KN)$ by matrix searching. Later one-dimensional clustering treatments extend the cost structures and discuss reconstruction and space reductions (Grønlund et al., 2017). Thus contiguous-cell dynamic programming, continuous reproduction-point optimization, and an $O(mk)$ matrix-search consequence are not new as generic claims. In the present deterministic problem, participation forces a cell’s reconstruction to its smallest cap, not its least-squares centroid. Its loss includes branch-specific intermediate-service costs. The difference is a particular feasible distortion, not a new ordered-partition paradigm.

Unbiased randomization and dual quantization. The adjacent lottery is exactly the classical unbiased two-point quantizer. For distinct adjacent levels $u < v$ and $u \leq b \leq v$,

$$\Pr(C = v) = \frac{b - u}{v - u}, \quad \Pr(C = u) = \frac{v - b}{v - u}, \quad \mathbb{E}C = b. \quad (1)$$

Stochastic rounding uses this same rule (Crocì et al., 2022). Dual quantization is an even closer optimization predecessor: Pagès and Wilbertz (2012) optimize mean-preserving random splitting and relate it to piecewise-affine interpolation. Their extended formulation also uses nearest-neighbor projection outside the grid’s convex hull. We therefore do not claim that random splitting, optimized grids, interpolation, or treatment of out-of-hull points first arise here.

For the quadratic terminal reward, our in-hull loss is $q(b - u)(v - b)/2$, the corresponding quadratic dual-quantization error multiplied by $q/2$. Above the highest level z , however, the contractual loss is

$$f(b) - f(z) + h_j(b - z) = (r - qz)(b - z) + \frac{\gamma_j - q}{2}(b - z)^2.$$

It is generally not a common squared-distance projection penalty: it carries a level-dependent linear term and a heterogeneous intermediate-service curvature. The paper’s combined structural result concerns this one-sided contractual tail, its continuously chosen endpoint, and the cancellation that preserves the terminal Monge order. The affine movement of an interior codeword follows the quadratic interpolation structure and is not advertised as an independent general quantization principle. Its sufficient conditions are explicit: saturated branch targets, quadratic terminal reward, uncharged codewords, and intermediate costs arising only above the highest level. Section 8 treats endogenous targets, and Section 9 treats actual level charges

without assuming the same anchoring lemma.

Finite-rate control and contracts. An informed encoder and a downstream finite-alphabet decoder are standard in communication-constrained control (Fu, 2024). Our result is not a stabilization or channel-capacity theorem. There is no dynamic plant whose state is estimated through noisy feedback, no hidden type report, and no incentive-compatibility claim. What the contract adds is the allocation of accepted continuation payments and the distinction between conditional expected and realization-wise participation. Continuation promises themselves have a long contracting history (Spear and Srivastava, 1987; Thomas and Worrall, 1988), including finite policy-graph implementations (Zhang, 2012). Below the exact alphabet threshold, participation determines the allowable quantizer rather than communication alone determining the objective.

Classical matrix search, model-specific cost structure. Quadrangle-inequality dynamic programming (Yao, 1980) and full totally monotone matrix search (Aggarwal et al., 1987) supply the algorithmic framework. Generalized and partial Monge matrix searching require their own domain analysis (Klawe and Kleitman, 1990; Chan, 2021; Kaplan et al., 2017). Our suffix and prefix domains have the favorable completion orientation; this is not an improvement on the general partial-matrix bounds. We retain complete proofs of the finite tie rule, wholly padded selected rows, column offsets, and reachable predecessor intervals so that the implementation’s use of the classical result is checkable. The essential contractual input is the inequality after continuous tail minimization, not merely convexity of a scalar reproduction-point objective. We do not claim that minimizing a reproduction point preserves Monge structure for arbitrary quantizer losses.

Table 1 states the predecessor principle and contractual increment for each result.

3 The Contract and Its Execution Boundary

3.1 A saturated renewal agreement

At the renewal date, a public branch $j \in \{1, \dots, k\}$ is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. Its continuation cap is $0 < b_1 < \dots < b_k < 1$. The provider chooses an intermediate tier $Y_j \in [0, 1]$ and a terminal tier $C_j \in [0, 1]$. All branches subsequently reach the same terminal public vertex. Unit payment coefficients and unit discounting make total remaining payment $Y_j + C_j$. A fixed service-benefit continuation minus the customer’s outside continuation gives the cap. At the root the provider has committed to

$$\sum_j \pi_j \mathbb{E}[Y_j + C_j] = \bar{b}, \quad \bar{b} := \sum_j \pi_j b_j. \quad (2)$$

Under expected participation, $\mathbb{E}[Y_j + C_j] \leq b_j$ for every j , with expectation conditional on that observed branch and taken before the internal draw. Positivity of the probabilities and (2) imply equality separately on every branch. Under pathwise participation, the stronger constraint is $Y_j + C_j \leq b_j$ almost surely. The same argument then implies equality almost surely, not merely in expectation.

The operating payoff is $f(C_j) - h_j(Y_j)$, with increasing concave f and convex nondecreasing h_j satisfying $h_j(0) = 0$. General shape assumptions suffice for the deterministic and pathwise results. The global continuous algorithm specializes to the quadratic primitives in (8). The negative intermediate contribution represents the net cost of an advance or expedited tier, while the terminal tier contributes net operating return. It does not mean that the customer has negative service benefit: that benefit has already entered the continuation cap. There are no endogenous public transitions, coupled commitments, or switching costs in the renewal theorem.

3.2 What a writable code means

At the observed branch, an encoder selects a symbol $Z \in \{1, \dots, m\}$. At the terminal vertex the decoder observes only the current public vertex and Z . Its stored program can contain arbitrary read-only model constants but cannot recover the realized branch from those constants. An inherited intermediate tier, a customer identifier with a branch lookup, a retained shared random seed, and a transmitted numerical payment are all additional history channels; none is free in this interface. An m -symbol alphabet requires $\lceil \log_2 m \rceil$ fixed-width writable bits, not m bits. Public-state storage and read-only program storage are counted separately.

Private encoder randomization is allowed under expected participation. Fresh decoder randomness is independent of earlier private choices conditional on the symbol. Any seed that correlates those choices across dates must itself be encoded in the charged symbol. Concavity permits a deterministic terminal decoder at the expected-participation optimum, so a codebook is a collection of terminal levels. The lower bound for exact implementation continues to allow randomized encoders and decoders under this accounting. Distinct full-information terminal actions and strict concavity remain visible assumptions; Proposition 8.2 handles repeated caps; uniqueness is required for the exact-action lower bound.

3.3 Acceptance timing

The renewal desk observes the branch and offers the intermediate tier and terminal lottery. Under expected participation, the customer accepts before the ticket draw; after acceptance only the ticket reaches the common gateway. The realized ticket may be disclosed, but no new costless exit option is supplied after the draw within this episode. Under pathwise participation, each possible realized obligation must instead satisfy the branch cap, and a draw-specific intermediate tier is allowed. A seed known before acceptance changes that conditioning information. Repeated-customer effects require an enlarged state and are not part of the three-date theorem. The full stipulated mechanism and its qualifications are given in the electronic companion; no empirical or legal claim about an actual provider is made. Section 8 allows a nonsaturated root promise, and Section 9 specifies the intermediate risk protocol.

4 The Deterministic Service-Code Frontier

Consider three dates. The root tier is fixed at zero. At the intermediate date a branch j is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. All branches then enter the same terminal public vertex. Payments and discounting equal one. Intermediate and terminal tiers are in $[0, 1]$. Branch j has cap $0 < b_1 < \dots < b_k < 1$, and the root promise is $\bar{b} = \sum_j \pi_j b_j$. At the intermediate date the reward is $-h_j(y)$, where h_j is convex, nondecreasing, and $h_j(0) = 0$. The common terminal reward $f(c)$ is continuous, concave, and strictly increasing on $[0, 1]$. The terminal cap is one. The terminal decoder observes only its public vertex and its memory symbol, not the inherited intermediate tier.

Theorem 4.1 (Heterogeneous renewal frontier). *The unique full-information policy is $y_j = 0$, $c_j = b_j$, with value $J = \sum_j \pi_j f(b_j)$. Exact implementation requires k symbols under strict cap ordering; repeated caps are covered by Proposition 8.2. With at most m deterministic terminal symbols, an optimal encoding has contiguous nonempty cells in cap order. For a cell $i, \dots, j$, its terminal tier is b_i and its loss is*

$$C(i, j) = \sum_{h=i}^j \pi_h [f(b_h) - f(b_i) + h_h(b_h - b_i)]. \quad (3)$$

The exact minimum loss is $D_m = D(\min(m, k), k)$, where

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (4)$$

and all other infeasible entries equal $+\infty$. With a precomputed cell-cost table this requires $O(mk^2)$ arithmetic operations. For quadratic f and quadratic h_j , prefix sums of the probability-weighted coefficients compute each cell cost in constant arithmetic time.

The proof is in Electronic Companion Section EC.1.1.

This strengthens and includes the quadratic equal-probability frontier. With $f(c) = 2c - c^2/2$, $h_j(y) = y^2/2$, and $\pi_j = 1/k$, (3) becomes

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (5)$$

For $k = 8$ and $b_j = j/9$, one, two, and three symbols lose $119/162$, $5/18$, and $49/324$, respectively. Two writable bits permit four symbols, not three. The segmentation recurrence is standard; the contract-specific cell loss, ordered-cell sufficiency, and exact memory interpretation are the substantive statements.

5 When Randomization Improves a Restricted Memory Budget

The exact-optimum lower bound in Theorem 8.1 does not imply that randomization is useless below that alphabet size. The renewal family permits an explicit answer under the declared conditional-expectation participation institution. At an intermediate public branch, the controller may randomize its terminal symbol;

it carries no uncharged seed or branch record. Participation is evaluated before that private draw. Terminal actions may depend only on the terminal public vertex and the retained symbol.

For terminal levels $c_1 < \dots < c_s$, let F_c be the linear interpolant through $(c_i, f(c_i))$. Since f is concave and strictly increasing, this interpolant is concave and strictly increasing on $[c_1, c_s]$; for $s = 1$ it is defined at its sole point. Retain the caps, probabilities, and convex intermediate costs of Theorem 4.1.

Theorem 5.1 (Optimal lotteries for a fixed terminal alphabet). *For a fixed set of terminal levels, a feasible randomized controller exists if and only if $c_1 \leq b_1$. Its optimum at branch j has deterministic intermediate tier and terminal mean*

$$\mu_j = \min\{b_j, c_s\}, \quad y_j = b_j - \mu_j.$$

It randomizes over at most the two adjacent terminal levels bracketing μ_j . The optimal loss with at most m writable terminal symbols is consequently

$$D_m^{\text{ran}} = \min_{\substack{1 \leq s \leq m, \\ 0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq b_1}} \sum_j \pi_j \{f(b_j) - F_c(\min\{b_j, c_s\}) + h_j((b_j - c_s)_+)\}. \quad (6)$$

The minimum allows coincident levels by deleting unused duplicates. It satisfies $D_m^{\text{ran}} \leq D_m$. Equality with the full-information optimum still requires k symbols when the full-information actions are distinct and rewards are strictly concave. Formula (6) optimizes the encoding for each continuous codebook; it is not a claim that the joint codebook problem is solved by the deterministic segmentation recurrence.

The proof is in Electronic Companion Section EC.1.2.

Proposition 5.2 (A strict, exactly solved two-symbol separation). *In the quadratic renewal model, take three equiprobable branches with caps $1/4, 1/2, 3/4$, $f(c) = 2c - c^2/2$, and $h_j(y) = y^2/2$. Then*

$$D_2 = \frac{1}{8}, \quad D_2^{\text{ran}} = \frac{1}{96}, \quad D_3 = D_3^{\text{ran}} = 0. \quad (7)$$

Two symbols thus reduce the randomized loss to one twelfth of the best deterministic loss, without reducing the three-symbol requirement for exact implementation.

The proof is in Electronic Companion Section EC.1.3.

The institutional qualification is substantive. On the middle branch, the high private draw delivers terminal payment $3/4$ against an expected-continuation cap $1/2$. The lottery is feasible when participation is checked before the private draw, as stipulated here, but not when that branch cap must hold for every private draw. We do not transfer the randomized frontier to the pathwise institution. The deterministic frontier and the exact-memory lower bound retain their original domains. This separation explains why a statement about the minimum alphabet at the exact optimum cannot settle the economic value of randomization under a smaller alphabet.

6 The Global Continuous Lottery Codebook

The fixed-codebook formula in Theorem 5.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem 8.1.

Retain the three-date renewal architecture of Theorem 4.1, including the saturated root promise $\bar{b} = \sum_{j=1}^k \pi_j b_j$, strictly ordered caps $0 < b_1 < \dots < b_k < 1$, and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (8)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when $r = q$. All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

6.1 Cap anchors and one continuous terminal level

For $u \leq b \leq v$, quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (9)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

Lemma 6.1 (Cap-anchored representation). *For every alphabet budget $m \geq 1$, an optimum of (6) exists with the following representation. Its lowest codeword is b_1 ; every codeword except the highest belongs to $\{b_1, \dots, b_k\}$; and the highest belongs to $[b_1, b_k]$. A one-symbol optimum is $\{b_1\}$. The highest codeword need not be a branch cap.*

The proof is in Electronic Companion Section EC.1.4.

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition 6.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

6.2 A finite exact algorithm

For adjacent anchored levels $b_i < b_j$, define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (10)$$

Let $P(s, j)$ be the minimum interpolation loss on caps through b_j using exactly s anchored levels, starting at b_1 and ending at b_j . Set $P(1, 1) = 0$, $P(1, j) = +\infty$ for $j > 1$, and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (11)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma 6.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that b_i is the last anchored level and z is the final level. For $\ell \geq i$ and $b_\ell \leq z \leq b_{\ell+1}$, put $b_{k+1} = b_k$ and define

$$\begin{aligned} \Psi_{i\ell}(z) = & \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ & + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (12)$$

The first sum is the lottery loss below z ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For $\ell < k$ the function is strictly convex in z and has the constrained minimizer

$$z_{i\ell} = \left[\frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (13)$$

For $\ell = k$, take $z_{ik} = b_k$. Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (14)$$

Theorem 6.2 (Global frontier and exact complexity). *Under (8), the optimum with at most m symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (15)$$

where the inner minimum is empty for $m = 1$. Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and $1 \leq m \leq k$, all frontiers through budget m can be computed in $O(mk^2)$ exact rational operations and $O(mk + k)$ stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length L , a reduced-rational implementation has polynomial Turing complexity. A conservative bound is $O(mk^2 S^3)$ bit operations with $S = O(k[L + \log(k+2)])$. This includes exact cost comparisons, clamping, and fraction reduction. For $m \geq k$ the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (6).

The proof is in Electronic Companion Section EC.1.5.

6.3 A continuous optimal highest level

Proposition 6.3 (An off-cap terminal level is optimal). *Let $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$, $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$, $r = 2$, $q = 1$, and $\gamma_j = 1$. The globally optimal two-symbol expected-participation codebook is $(1/4, 5/8)$, and*

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (16)$$

where the cap-only minimum is over feasible codebooks and $D(c)$ denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

The proof is in Electronic Companion Section EC.1.6.

6.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write $D_m^{\text{pw,ran}}$ for the best randomized loss under these pathwise constraints and the same saturated root promise.

Theorem 6.4 (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem 4.1, with strictly increasing concave f and convex nondecreasing h_j , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (17)$$

Consequently (15) and the deterministic frontier give an exact comparison of the two participation institutions under (8). Their exact-optimum alphabet thresholds are both k when strict concavity makes the k distinct full-information actions unique.

The proof is in Electronic Companion Section EC.1.7.

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem 6.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

7 Monge Costs and Classical Linear Matrix Search

The finite dynamic program still optimizes a continuous highest service level. We first prove the contractual cost inequalities and then verify the classical easy-direction completion with the exact order used by the code. A search routine cannot establish the economic reduction that supplies its entries.

7.1 The inequality survives continuous terminal minimization

Write $I_\ell = [b_\ell, b_{\ell+1}]$, including $I_k = \{b_k\}$, and $Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z)$ for $i \leq \ell$. A partial array is Monge here when

$$A(r, c) + A(s, d) \leq A(r, d) + A(s, c), \quad r < s, c < d, \quad (18)$$

whenever all four entries are defined. This fixes the inequality convention before any orientation is named.

Lemma 7.1 (Contractual finite-cost inequalities). *The anchored edge costs E and deterministic cell costs C satisfy $A(a, c) + A(b, d) \leq A(a, d) + A(b, c)$ for $a \leq b \leq c \leq d$. The continuously minimized terminal array Q is Monge on its feasible domain. These statements permit arbitrary positive probabilities and heterogeneous nonnegative intermediate curvatures.*

The proof is in Electronic Companion Section EC.1.8.

Lemma 7.2 (Previous-layer values are column offsets). *Adding any finite column term v_c preserves the Monge inequality. For fixed finite predecessor values, the arrays $A(j, i) = P(s, i) + E(i, j)$ and $B(j, i) = D(s - 1, i) + C(i + 1, j)$ are therefore Monge wherever all four entries are feasible.*

The proof is in Electronic Companion Section EC.1.9.

7.2 Completion with the implemented strict and tie orders

Give a feasible entry the lexicographic key

$$K(r, c) = (0, A(r, c), c). \quad (19)$$

Finite ties select the smaller column, and distinct columns have distinct keys. The exact total-monotonicity condition used by the implementation is

$$K(r, d) < K(r, c) \implies K(s, d) < K(s, c), \quad r < s, c < d. \quad (20)$$

A strict preference for the right column cannot reverse in a later row. This applies to arbitrary selected rows and columns. Because keys are distinct, a decrease of row-argmin indices in any submatrix would contradict (20).

Lemma 7.3 (Suffix-feasible completion). *Let row r have feasible columns $c \geq \ell_r$, with nondecreasing cutoffs ℓ_r . Suppose the finite entries are Monge. Complete every $c < \ell_r$ with $K(r, c) = (1, 0, -c)$. Then (20) holds. A completely infeasible row of a selected submatrix selects its largest selected column.*

The proof is in Electronic Companion Section EC.1.10.

Lemma 7.4 (Prefix-feasible completion). *Let row r have feasible columns $c \leq u_r$, with nondecreasing cutoffs u_r . Suppose the finite entries are Monge. Complete every $c > u_r$ with $K(r, c) = (1, 0, c)$. Then (20) holds. A completely infeasible selected row selects its smallest selected column.*

The proof is in Electronic Companion Section EC.1.11.

These lemmas specialize the known favorable completion setting in Section 2; they are not general partial-matrix search theorems. A helper accepting two bounds does not prove total monotonicity for arbitrary interval rows. For example, two zero-cost rows feasible only at columns 2 and 1 have a vacuously Monge finite part but decreasing minima. Uniform infinite padding with ordinary left ties also fails on the required suffix domain: rows with cutoffs 2, 3 restricted to columns 1, 2 select 2, 1. The opposite internal padding orders above are deliberate.

7.3 Reachability and three search interfaces

Lemma 7.5 (Contiguous reachable predecessor sets). *For the anchored program, $P(1, j)$ is finite exactly at $j = 1$, and for $s \geq 2$, $P(s, j)$ is finite exactly when $s \leq j \leq k$. For the deterministic program, $D(0, j)$ is finite only at $j = 0$, and $D(s, j)$ is finite exactly when $s \leq j \leq k$ for $s \geq 1$. Each globally reachable predecessor set that can extend to a further endpoint is a contiguous interval or the initial singleton.*

The proof is in Electronic Companion Section EC.1.12.

The following pseudocode uses one-based cap indices and zero for the empty deterministic prefix. “Search” means ordinary full-matrix SMAWK on the supplied key. All prior-layer values are known before a search.

Terminal interface. Rows and columns are $1, \dots, k$. At (i, ℓ) return $(1, 0, -\ell)$ if $\ell < i$; otherwise minimize (12) by (13) and return $(0, Q(i, \ell), \ell)$. Search all rows. Reevaluate each winning cell once to store T_i and its continuous optimizer.

Anchored-prefix interface. To form layer $s+1$, use rows $j = s+1, \dots, k$. Columns are $\{1\}$ for $s = 1$, and $s, \dots, k-1$ otherwise. At (j, i) return $(1, 0, i)$ if $i \geq j$, and $(0, P(s, i) + E(i, j), i)$ otherwise. Search and store values and predecessors. Scan the current reachable endpoints with T_i to update the at-most-budget incumbent; backtrack only its winning codebook.

Deterministic-prefix interface. To form layer s , use rows $j = s, \dots, k$. Columns are $\{0\}$ for $s = 1$, and $s-1, \dots, k-1$ otherwise. Return $(1, 0, i)$ if $i \geq j$, and $(0, D(s-1, i) + C(i+1, j), i)$ otherwise. Search, store predecessors, and backtrack the solution ending at k .

Lemma 7.5 justifies every discarded column. The terminal interface has shrinking feasible suffixes; the program interfaces have expanding feasible prefixes. The completion lemmas include infeasible rows in submatrices created internally by SMAWK, although every original searched row has a feasible entry. Padded comparisons do not evaluate an undefined economic cost.

Theorem 7.6 (Contractual corollary of classical matrix search). *For ordered rational inputs satisfying (8) and $1 \leq m \leq k$, both expected-participation and deterministic/pathwise frontiers through budget m , including an optimal codebook for every budget, require $O(mk)$ rational operations and comparisons and $O(mk + k)$ stored rational numbers and indices. The highest randomized level remains continuous. With total input length L and $S = O(k[L + \log(k+2)])$, a conservative Turing bound is $O(mkS^3)$. Sorting unordered distinct caps adds $O(k \log k)$ exact comparisons.*

The proof is in Electronic Companion Section EC.1.13.

The retained divide-and-conquer implementation is a valid $O(mk \log(k+1))$ alternative using ordinary row-minimum monotonicity. It is an implementation comparator rather than another headline theorem. $O(k)$ exact-rational oracle/comparison work per layer does not imply uniform wall-time superiority for a particular Python implementation.

8 Promise Allocation Away from Saturation

The saturated promise is an endpoint of a larger allocation problem, not a normalization. Fix the branch caps and replace the root promise by $B \in [0, \bar{b}]$, where $\bar{b} = \sum_j \pi_j b_j$. Write $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for optimal value with at most m symbols under expected participation, pathwise participation allowing draw-specific intermediate tiers, and deterministic execution, respectively. The information interface remains unchanged. There is no codeword-dependent charge in this section, and terminal levels may range continuously over $[0, 1]$. The full-information value is $V_\infty(B)$.

8.1 An exact threshold for every root promise

Theorem 8.1 (Clipped service levels and exact alphabet). *Suppose f is strictly increasing and strictly concave, and each h_j is convex, nondecreasing, with $h_j(0) = 0$. For every $B \in [0, \bar{b}]$, full information uniquely implements*

$$Y_j = 0, \quad C_j = \min\{b_j, \tau\}, \quad \sum_j \pi_j \min\{b_j, \tau\} = B. \quad (21)$$

At $B = 0$ take $\tau = 0$, and at $B = \bar{b}$ take $\tau = \max_j b_j$. The minimum alphabet attaining this value under any of the three institutions is

$$n(B) = |\{\min(b_j, \tau) : 1 \leq j \leq k\}|.$$

In particular, for $0 \leq B \leq \min_j b_j$, one symbol attains $V_m^I(B) = f(B)$ for every $m \geq 1$ and $I \in \{E, P, D\}$.

Proof. For a full-information feasible policy, let $t_j = \mathbb{E}(Y_j + C_j) \leq b_j$. Replacing it on branch j by $Y_j = 0, C_j = t_j$ preserves the root promise and satisfies even pathwise participation. Jensen's inequality, strict increase of f , and $h_j \geq 0$ show that the replacement weakly improves value, strictly if $\mathbb{E}Y_j > 0$ or C_j is nonconstant. Hence it suffices to maximize $\sum_j \pi_j f(t_j)$ with $0 \leq t_j \leq b_j$ and $\sum_j \pi_j t_j = B$. Strict concavity implies uniqueness. If two uncapped coordinates differ, transferring equal probability-weighted amounts from the larger to the smaller strictly improves this sum; the same transfer shows that a capped coordinate below the common free level must be filled. Thus the unique optimizer is (21), with the endpoint cases interpreted as stated. Continuity of the clipped mean gives existence of τ .

Equality in these improvements requires deterministic terminal action $\min(b_j, \tau)$ on each branch. A symbol's terminal decoder cannot implement two different deterministic actions at the same public vertex. Its conditional output distribution is independent of the originating branch, including when fresh decoder randomization is permitted. Consequently distinct optimal actions require distinct symbols. Assigning one

symbol per distinct action attains the lower bound under all institutions. When $B \leq \min b_j$, all optimal actions equal B . $\square$

For strictly ordered caps, the thresholds at which $n(B)$ changes are explicit: put $B_\ell = \sum_j \pi_j \min(b_j, b_\ell)$. The exact alphabet is $\ell + 1$ for $B_\ell < B \leq B_{\ell+1}$, $1 \leq \ell < k$, and it is one on $[0, B_1]$. At $B = B_\ell$ it is ℓ . Thus exact implementation has a promise-dependent phase diagram, rather than requiring k symbols at every promise.

Proposition 8.2 (Repeated caps at the saturated endpoint). *For $B = \bar{b}$, branches with a common cap b can be aggregated in every deterministic, expected, and pathwise frontier. Their aggregate probability is $p_b = \sum_{j:b_j=b} \pi_j$ and their aggregate intermediate cost is $h_b(y) = p_b^{-1} \sum_{j:b_j=b} \pi_j h_j(y)$. Quadratic costs therefore aggregate by their probability-weighted curvature. The exact alphabet threshold is the number of distinct caps. For arbitrary B , the full-information threshold remains (21), but aggregation of the restricted-alphabet problem with heterogeneous intermediate costs is not asserted.*

Proof. At saturation, the fixed-book expected rule chooses the same terminal mean and intermediate tier for all branches with the same cap. The deterministic and pathwise rules choose the same highest eligible terminal level. Summing their rewards gives precisely the aggregate cost. Optimize over the common books and apply Theorem 8.1 for the threshold. Away from saturation the branch promises themselves can differ, so the saturated aggregation argument does not apply. $\square$

8.2 Uniform transport between any two root promises

Assume in addition that f is Lipschitz with constant L_f on $[0, 1]$ and each h_j is Lipschitz with constant H_j on $[0, b_j]$; put $H = \max_j H_j$. These constants are independent of the alphabet budget. For quadratic primitives one may take $L_f = r$ and $H = \max_j \gamma_j b_j$. The smaller intermediate domain is sufficient because the mean and pathwise replacements used below have tiers at most b_j .

Theorem 8.3 (All-promise value and institutional-gap stability). *For any $0 \leq B_1 \leq B_2 \leq \bar{b}$, $\Delta = B_2 - B_1$, every $m \geq 1$, and $I \in \{E, P, D\}$,*

$$V_m^I(B_2) - L_f \Delta \leq V_m^I(B_1) \leq V_m^I(B_2) + H \Delta. \quad (22)$$

If $G_m(B) = V_m^E(B) - V_m^P(B)$, then

$$|G_m(B_2) - G_m(B_1)| \leq (L_f + H) \Delta. \quad (23)$$

The same bound holds for changes in the loss $V_\infty(B) - V_m^I(B)$. In particular, a certified strict advantage $G_m(B_0) > 0$ persists for every feasible B with $|B - B_0| < G_m(B_0)/(L_f + H)$.

Proof. First contract any policy at $B_2 > 0$ by multiplying all intermediate and terminal actions by $\alpha = B_1/B_2$. The decoder uses the same symbols, and both participation conventions remain feasible. Intermediate cost weakly decreases. The decrease in terminal reward is at most $L_f(1 - \alpha)\mathbb{E}C \leq L_f \Delta$, where expectations include branch weights. The case $B_2 = 0$ is immediate. Taking suprema gives the left inequality.

For the other direction, start from a policy at B_1 . Under expected participation, replace the terminal decoder given a symbol by its mean and the intermediate tier on each branch by its mean. These changes weakly improve reward, preserve the alphabet, and preserve expected obligations. Write $\mu_j = \mathbb{E}C_j$, $y_j = \mathbb{E}Y_j$, and $s_j = b_j - y_j - \mu_j \geq 0$. If $B_1 < \bar{b}$, put $\theta = \Delta/(\bar{b} - B_1)$ and replace y_j by $y_j + \theta s_j$. The total added obligation is $\theta \sum_j \pi_j s_j = \Delta$. Its cost is at most $H\Delta$, and the book and its encoding probabilities are unchanged.

For pathwise participation, replace fresh terminal decoder randomness conditional on symbol z by its mean c_z . Conditional independence from the earlier tier ensures $Y_j + c_z \leq b_j$ almost surely after this replacement; terminal concavity weakly improves reward. Use the draw-specific tier

$$Y'_j = Y_j + \theta(b_j - Y_j - c_z).$$

This is admissible at the stipulated intermediate timing, remains between Y_j and $b_j - c_z$, and adds expected obligation Δ with cost at most $H\Delta$. It requires no extra information at the terminal gateway. The deterministic case is the same construction without random draws. If $B_1 = \bar{b}$, necessarily $\Delta = 0$. Taking suprema, or limits of near-optimal policies, proves the right inequality without requiring a maximizer.

For each institution the value increment lies in $[-H\Delta, L_f\Delta]$. Subtracting two such increments proves the institutional-gap bound. The same constructions apply with unrestricted full information, yielding the loss bound. Every transformation preserves the original number of writable symbols; no concavity of the optimized codebook envelope is assumed. $\square$

In the equal-probability three-branch example of Proposition 5.2, $\bar{b} = 1/2$, $G_2(\bar{b}) = 11/96$, and $L_f + H = 2 + 3/4 = 11/4$. Therefore

$$G_2(B) \geq \frac{11}{96} - \frac{11}{4} \left(\frac{1}{2} - B \right) > 0 \quad \text{for } \frac{11}{24} < B \leq \frac{1}{2}. \quad (24)$$

This is a certified interval of strict advantage, not a numerical continuation or a claim of an exact nonsaturated loss formula. At the other endpoint, $B \leq 1/4$ gives zero loss with one symbol. In between, cap anchoring at the original b_j and pathwise–deterministic equality do not follow from the saturated proofs. Indeed the one-symbol full optimum $C = B < \min b_j$ already shows why original-cap anchoring cannot hold for all promises. The decision insight persists without implying that the saturated algorithm solves every interior promise.

8.3 The nonsaturated allocation problem

The larger problem has an exact reduction even when it does not have the saturated finite-anchor reduction. For a fixed book $c = (c_1, \dots, c_s)$ let F_c be its concave interpolant and define

$$v_{j,c}(t) = F_c(\min\{t, c_s\}) - h_j((t - c_s)_+), \quad c_1 \leq t \leq b_j. \quad (25)$$

Proposition 8.4 (Joint promise allocation and codebook design). *The expected-participation value at any B*

is exactly

$$V_m^E(B) = \max_{\substack{|c| \leq m, 0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq t_j \leq b_j, \sum_j \pi_j t_j = B}} \sum_j \pi_j v_{j,c}(t_j). \quad (26)$$

For a fixed book this is a separable concave resource-allocation problem. Branch targets are the chosen t_j , not mechanically the original caps. Saturation fixes $t_j = b_j$ and recovers the earlier frontier.

Proof. Set $t_j = \mathbb{E}(Y_j + C_j)$ for an arbitrary feasible policy. The mean replacements in Theorem 5.1 are still valid. For a given t_j , maximizing $F_c(\mu) - h_j(t_j - \mu)$ over $c_1 \leq \mu \leq \min(t_j, c_s)$ selects its largest feasible mean by monotonicity. Conversely that mean and its adjacent lottery implement (25) with intermediate tier $(t_j - c_s)_+$. This proves both directions of (26). The response is concave: to the left of c_s it is F_c , to the right it is $f(c_s) - h_j(t - c_s)$, and the one-sided slopes drop from a nonnegative chord slope to a nonpositive cost slope. This also covers a singleton book. Allocation with a common supporting multiplier, allowing subgradient intervals at kinks, solves the fixed-book problem. No interchange of codebook optimization and a dual minimization is used. $\square$

Electronic Companion Proposition EC.6.1 gives an exact supporting-price algorithm and a separately checkable certificate for every fixed-book allocation. Exhaustive certified catalog search extends that computation to all promises on small catalogs; its exponential cost is stated explicitly.

9 Joint Design of a Risk-Limited Certified Alphabet

A finite certification catalog gives a general intermediate institution and a nontrivial codebook-dependent implementation problem. Let $\mathcal{A} = \{a_1 < \dots < a_N\} \subset [0, 1]$ be the allowable terminal levels, with $a_1 \leq b_1$. Selecting a level a incurs an arbitrary nonnegative charge $\rho(a)$; for example, it may combine a declared precision/storage charge and a per-level certification charge in the same units as service loss. These are stipulated prices, not estimated operational costs. A book is any subset of $\mathcal{A}$ of size at most m . The objective includes the charges of its actual selected levels, not merely its declared capacity.

The root promise is saturated in this section. Intermediate service y_j is chosen before the symbol draw, with

$$y_j + \mathbb{E}C_j = b_j, \quad y_j + C_j \leq b_j + \delta \quad \text{almost surely.} \quad (27)$$

This explicitly defined institution has the two familiar endpoints: $\delta = 0$ attains the pathwise/deterministic catalog optimum, while $\delta \geq 1$ removes the overrun restriction. The pre-draw restriction is not asserted without loss for arbitrary intermediate risk protocols.

For adjacent selected levels $u < v$, assign to this edge the branches $u \leq b_j < v$ and define

$$K_\delta(u, v) = \sum_{j: u \leq b_j < v} \pi_j \ell_j^\delta(u, v), \quad (28)$$

where, on that interval,

$$\ell_j^\delta(u, v) = \begin{cases} f(b_j) - \frac{v - b_j}{v - u} f(u) - \frac{b_j - u}{v - u} f(v), & v - b_j \leq \delta, \\ f(b_j) - f(u) + h_j(b_j - u), & v - b_j > \delta. \end{cases} \quad (29)$$

At a selected cap the branch loss is zero, including the boundary $b_j = u$. The last selected level u carries tail cost

$$T(u) = \sum_{j: b_j \geq u} \pi_j \{f(b_j) - f(u) + h_j(b_j - u)\}. \quad (30)$$

No original-cap anchoring is imposed on catalog entries.

Theorem 9.1 (Global catalog frontier with level-dependent charges). *For strictly increasing concave f and convex nondecreasing h_j , the following dynamic program globally minimizes service loss plus selected-level charges under the bounded-overflow institution:*

$$\begin{aligned} H_1(v) &= \begin{cases} \rho(v), & v \leq b_1, \\ +\infty, & v > b_1, \end{cases} \\ H_s(v) &= \rho(v) + \min_{u \in \mathcal{A}: u < v} \{H_{s-1}(u) + K_\delta(u, v)\}, \quad s \geq 2, \\ W_m^\delta &= \min_{1 \leq s \leq \min(m, N), v \in \mathcal{A}} \{H_s(v) + T(v)\}. \end{aligned} \quad (31)$$

Backtracking recovers the book, intermediate tiers, and lottery probabilities. All budgets through m require $O(kN^2 + mN^2)$ arithmetic/evaluation work with a direct branch-sum precomputation, and $O(N^2 + mN)$ storage. For rational quadratic data and rational catalog charges the calculation is exact in rational arithmetic. An arbitrary additional charge depending only on s can be added at the final minimization, but level charges must enter the recurrence itself.

Proof. For a fixed book, Proposition EC.5.1 gives the optimal branch rule. On $u \leq b_j < v$, if $v - b_j \leq \delta$ use the adjacent unbiased lottery of mean b_j and set $y_j = 0$; otherwise use u deterministically and set $y_j = b_j - u$. Branches above the last level also use that last level deterministically. Consequently the book's exact loss is the sum of its adjacent-edge costs and its terminal tail. Every branch is counted exactly once; cap endpoints contribute zero regardless of the interval convention.

Increasing paths through the catalog enumerate every possible sorted book. A path can begin exactly at levels no larger than b_1 , which is necessary and sufficient for feasibility. Charging ρ at each visited vertex adds the book's actual implementation cost. Induction on path length gives (31); appending the tail and minimizing over feasible lengths proves both inequalities needed for global optimality. If a selected level is unused, deleting it weakly lowers its nonnegative charge and does not worsen service. Thus the at-most-budget optimization also covers redundant symbols. Store predecessor indices and replay the fixed-book rule to reconstruct a controller.

There are $O(N^2)$ edges, each computable by k branch evaluations, and m layers of $O(N^2)$ comparisons/additions. Storing edges and backpointers gives the stated bounds. In the quadratic rational model

every interpolation, difference, and cost is rational. There is no continuous optimizer or rounding tolerance in the recurrence. At $\delta = 0$ every strictly bracketing upper level is excluded, so the branch rule is the deterministic one; at $\delta \geq 1$ every possible upper level is admissible. The endpoint claims follow after optimization over the same catalog. $\square$

This theorem solves a distinct, fully specified design problem for arbitrary k, N, m, δ and nonuniform ρ . It is not a discretization-based assertion of optimality over all real codewords, and it does not assume that risk-limited costs are Monge. The continuous uncharged expected frontier retains its separate $O(mk)$ guarantee; the certified-catalog theorem uses a general ordered-path algorithm to accommodate actual level charges and risk eligibility. Entropy coding, shared-table compression depending on pairs of levels, repeated-customer state, and arbitrary codebook-dependent costs are not silently reduced to additive ρ .

Why a loss frontier followed by a resource surcharge is insufficient. Use the off-cap model of Proposition 6.3 and catalog $\{1/4, 1/2, 5/8, 3/4\}$ with two symbols and $\delta = 1$. With zero charges, $(1/4, 5/8)$ attains loss $23/1280$. Give only level $5/8$ charge $1/640$. That book now costs $25/1280$, whereas each cap-only pair $(1/4, 1/2)$ and $(1/4, 3/4)$ costs $24/1280$. The optimum changes its selected levels, not merely its capacity. The exact catalog recurrence finds the change; retaining the uncharged loss-minimizing book and adding its storage bill does not.

Risk exposure along the continuous example. The original two-symbol continuous example is retained as an exact illustration beyond a fixed catalog. Its globally optimal top level and loss are

$$z_\delta = 1/2 + \min(\delta, 1/8), \quad D_2^\delta = z_\delta^2/20 - z_\delta/16 + 3/80. \quad (32)$$

On the middle branch the high-realization probability is

$$p_{\text{high}}(\delta) = \frac{1/4}{z_\delta - 1/4}, \quad \text{maximum overrun} = z_\delta - 1/2.$$

At zero tolerance this branch uses its cap deterministically; at tolerance $1/8$, the high probability is $2/3$ and the possible overrun is $1/8$. Thus probability, mean obligation, and maximum overrun are separate reported quantities. Figure 1 shows the continuous loss profile; the complete global-example proof is in the electronic companion.

10 Pricing Writable Capacity and Other Implementation Resources

Writable state, read-only constants, evaluation work, and compilation are different resources. A finite menu of implementations t for institution I , with charges depending only on capacity m , can be compared by

$$\min_{I,m,t} \{D_m^I + \lambda_w \lceil \log_2 m \rceil + \lambda_R R_{I,t}(m) + \lambda_T Q_{I,t}(m) + C_{I,t}(m)/H_0 + \kappa_I\}. \quad (33)$$

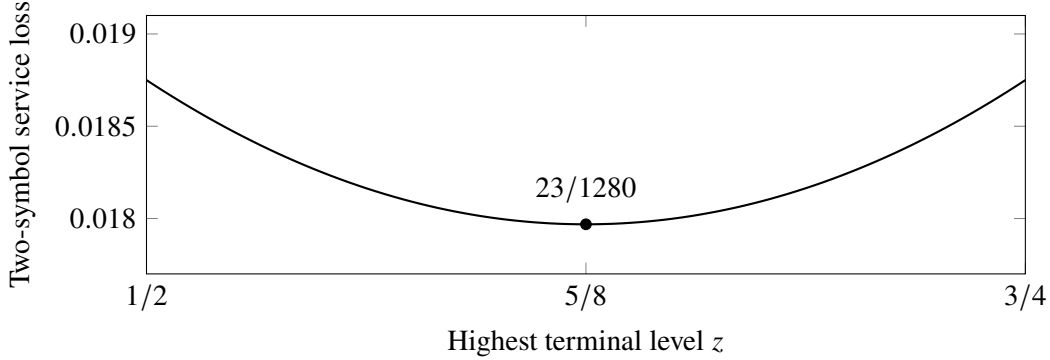


Figure 1: The analytically optimized highest level in the heterogeneous three-branch example. The lower level is $1/4$. The displayed function is $z^2/20 - z/16 + 3/80$ on $[1/2, 3/4]$; its interior minimum is $z = 5/8$, not a branch cap.

Here H_0 is the number of repeated deployments over which compilation is amortized, and κ_I is an exogenous protocol charge. This is accounting followed by finite comparison, not an additional structural optimization theorem. Its validity requires the stipulated menu charges to be independent of the particular book at a given capacity. Theorem 9.1, in contrast, puts actual additive level charges inside book selection.

With only a writable-bit price, capacity 2^w uses the frontier at $\min(2^w, k)$. For the equal-probability three-cap example, the zero-bit loss is $7/16$, the one-bit losses are $1/96$ under expected and $1/8$ under pathwise participation, and two bits attain full information. Thus a bit price strictly between $1/96$ and $1/8$ selects one bit under expected participation and two under pathwise participation when all other protocol charges agree. A different exogenous κ_I can change that comparison. This example interprets the proved frontiers; it does not estimate the monetary value of certification or customers' tolerance for risk.

11 Verification and Computational Evidence

The computational package separates exact finite checks, practical measurements, and a standard-quantizer comparison. The theorems establish the all-size and continuous claims; finite checks are regression evidence. Prices and branch populations are synthetic inputs, not empirical estimates.

New promise and catalog checks. Independent enumeration of all catalog subsets and all one/two-point branch supports agrees with every reported finite-catalog frontier, without assuming adjacent support in the comparison oracle. Tests include repeated caps, noncap catalog entries, zero intermediate curvature, nonuniform charges, and zero, intermediate, and unrestricted overrun tolerances. Clipped allocations satisfy exact weighted promises and allocation optimality conditions; direct scaling and saturation lifts verify the value inequalities on heterogeneous policies. Fixed-book responses are checked for concavity across their kinks. Invalid floating-point and malformed inputs are rejected.

The executed suite records 768 catalog frontier equalities, 26,604 direct-support book checks, 600 scaling/lift checks, 660 clipped-allocation checks, 2,460 response-concavity checks, and 128 repeated-cap

equalities. The charge-switch example and the interval in (24) are checked in exact fractions.

Comparison with an ordinary scalar quantizer. The baseline recursion over ordered cells is the same recursion used for scalar mean-square quantization; the cell oracle is not. For the three equal-probability caps, both optimal two-cell mean-square partitions have loss $1/96$. Their centroids are $(1/4, 5/8)$ or $(3/8, 3/4)$, respectively. Each merged cell includes a branch whose cap is below its centroid, so that centroid reconstruction is infeasible under the pathwise contract, even with nonnegative intermediate service. This exact comparison identifies the changed objective and constraint rather than benchmarking two algorithms that solve different problems. The expected contract instead uses the classical unbiased encoder with a contractual tail.

Inherited continuous-frontier evidence. The complete preceding exact suite, including completion ties, selected all-padded rows, controller replay, independent PADS substitution, and earlier regression suites, is rerun in isolated directories. Historical source and result files are not overwritten. The preceding process-isolated continuous-frontier timings through 16,384 branches and reduced-network HiGHS certificates are retained with their original provenance in the companion. The HiGHS comparison validates the finite reduction, not the cap-anchoring theorem or performance at the larger scale. PADS substitutes the search layer only. Neither comparison is presented as a new experiment on operational data.

Additional catalog landscapes. The executed catalog study uses uniform caps, two separated cap clusters, and a population with concentrated branch weights. Each family is evaluated at $(k, N) = (32, 16)$ and $(128, 32)$, eight-symbol capacity, and tolerances 0, $1/8$, 1, with heterogeneous curvatures and level prices. All 18 cases reconstruct and replay their selected books exactly. Times include catalog construction inside the solver, the dynamic program, reconstruction, and direct replay in one process; they are not directly compared with the predecessor’s process-isolated timings. Full numerical records and the timing definition accompany the code.

The nonsaturated extension additionally passes 1715 exact primal/dual-event comparisons, 6955 supporting branch maxima, 72 saturated catalog/DP equalities, 48 interior-promise catalog comparisons, and 480 arbitrary-promise transport checks. The supporting-price equality certifies each fixed-book solution; finite regression is not used as proof of the general theorem.

12 Conclusion

A renewal contract can make the distortion of a service code endogenous to participation and the allocation of continuation payments. The continuous saturated solution combines a classical unbiased quantizer inside its alphabet with a costly contractual tail above it. That tail permits an off-cap highest level while preserving the Monge order needed for classical linear-work matrix search. The expected-versus-pathwise distinction is a difference in feasible contracts, not a newly discovered randomization primitive.

The nonsaturated results give both a boundary map and a positive robustness conclusion: exact symbol requirements vary with the promise, low promises need no branch-specific terminal memory, and a strict

institutional advantage persists on an explicit interval below saturation. For finite certification catalogs, risk constraints and actual selected-level charges enter an exact joint design recurrence. These results distinguish continuous uncharged optimization, promise sensitivity, and charged catalog optimization without conflating their guarantees.

Data and code availability. All populations, prices, and curvatures in the experiments are synthetic and distributed with the implementation. The accompanying code contains an executable readme, exact input interfaces, seeded verification, raw measurements, and machine-readable certificates. One preparation command runs the current and historical exact tests in isolated output paths; a separate build command compiles committed sources and records every reader input and output hash. Historical measurements retain their original runner and timing definitions. No external customer data are needed. The repository reproduction instructions identify the current source revision and package paths without treating a generated PDF as its own source identity.

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Table 1: Theorem-level comparison with established principles.

Result here	Predecessor principle	Contract-specific statement proved here
Deterministic cells	Ordered quantizer DP; Wu	Participation fixes lower-cap reconstruction with heterogeneous intermediate costs.
Adjacent lottery	Stochastic rounding; dual quantization	Expected participation selects that primitive jointly with shortfall service; pathwise participation changes feasibility.
Continuous top level	Optimized quantizer grids and interpolation	Cap-anchored representation combined with a heterogeneous, one-sided contractual tail; exact off-cap optimum.
Linear frontier	Monge DP and classical matrix search	Tail crossing difference, optimized Monge order, and exact oracle/domain verification make the classical bound applicable.
Promise robustness	Concave allocation and Lipschitz comparison	Same-alphabet transport gives all-promise bounds, exact fixed-book certificates, and a strict nonsaturated advantage.
Charged risk catalog	Ordered shortest paths	Exact risk-eligibility edge costs and actual level charges jointly optimize service and implementation.